

캡스톤 디자인 9주차 발표

Capdol+I 고평채

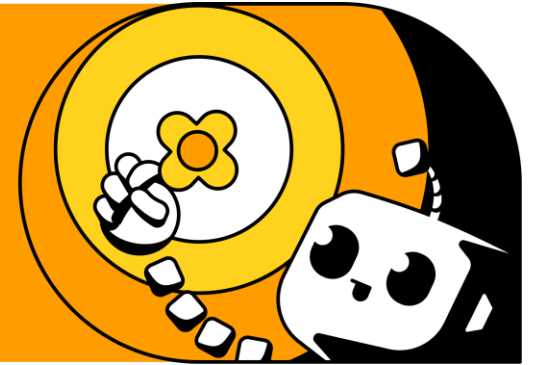
로봇팔 플랫폼 결정

오픈소스 소프트웨어



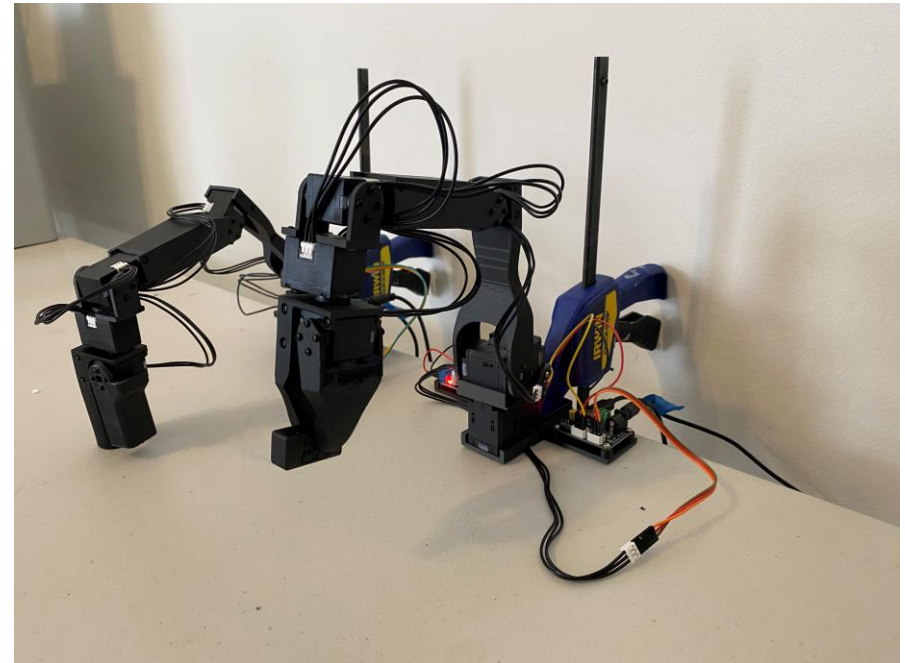
Hugging Face

LeRobot



오픈소스 하드웨어

<https://github.com/jess-moss/koch-v1-1>



서보모터 모델 결정

Feetech STS3215



45,800 KRW 1pcs
19.5kg·cm @6.0V

Robotis Dynamixel



VS

26,400 KRW 1pcs	52,800 KRW 1pcs
5.3kg·cm @5.0V	15.3kg·cm @12.0V

부속품

Bus Servo Adapter
– USB와 motor 사이 Serial 통신

5V power supply
– 5V 모터 전원 공급

Edge device

 Raspberry Pi 5



12V to 5V Voltage Reducer
– 5V, 12V 모터 동시 사용



12V power supply
– 12V 모터 전원공급

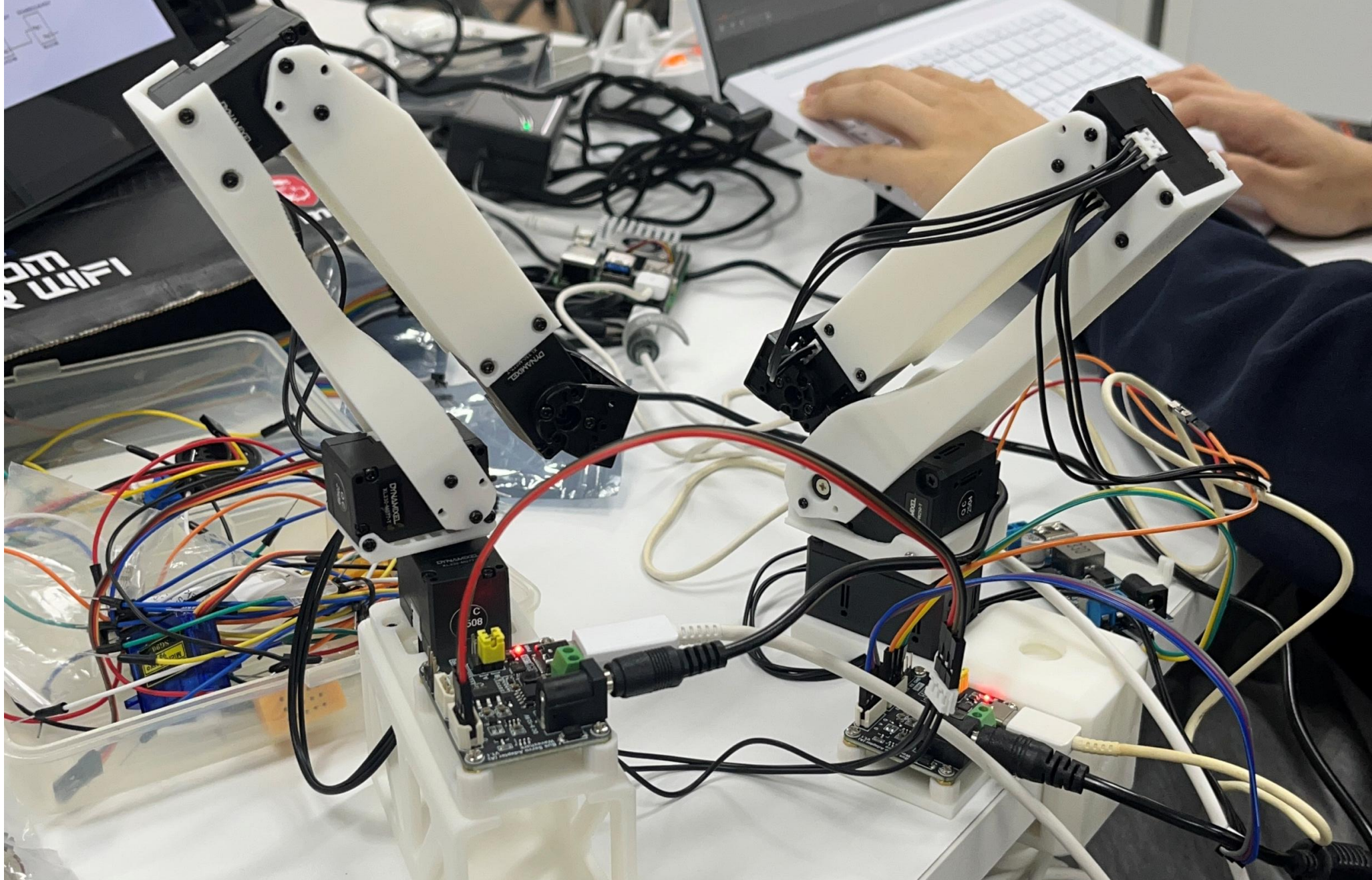


A photograph of a desk workspace during a robot assembly project. The desk is covered with various components, including a partially assembled white robot arm, several black Dynamixel XL330-M288-T servos, and their packaging. A red MSI MAG MORTAR motherboard is visible, along with a silver thermos, a white mouse, and a smartphone. The background shows a black office chair and a window.

최종 조립 사진

Follow arm

Reader arm



모터 인코더 값 읽기



트러블 슈팅

모터 초기불량

소프트웨어 상에서 토크를 주었을 때 무한 회전



원격동작 영상

